Gnan Reddy Bobba

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About

Aspiring Software Engineer having keen interest in building software systems.

Technical Skills

Languages: Go, Python, SQL
Backend: chi, PostgreSQL, Apache Kafka
Infrastructure: Docker, MinIO (S3)
Tools: Git, GitHub, Visual Studio Code

Projects

multimodal-rag-system Present
Go, Python, PostgreSQL (pgvector), Apache Kafka, MinIO (S3), Docker

- Building an MVP of an AI question-answering system (RAG) that ingests text and images and answers questions with source citations, using an event-driven pipeline.
- Currently it's under progress and is supervised by Dr. T. V. Gopal.

link | **sankomanga** Oct 2026
Go, AniList GraphQL API, HTML, CSS, JavaScript

- Built a manga search site as a single stdlib-only Go binary: a JSON backend over AniList's GraphQL API, with the HTML/CSS/JS frontend embedded in it.
- Added an in-memory TTL cache (**1,000**-entry cap) and a token-bucket limiter holding upstream calls to **20/min** under AniList's **30/min** limit, plus per-IP rate limiting and gzip compression.
- Backed by **75** tests run with `go test -race` and GitHub Actions CI on every push.

link | **webhook-delivery** Oct 2026
Go, PostgreSQL, chi, Docker

- Built a service in Go that reliably delivers event notifications (webhooks) to other apps, with signed requests and up to **8** retries with exponential backoff, using Postgres as its only infrastructure.
- Engineered per-endpoint isolation (concurrency caps, circuit breakers) so one failing endpoint can't stall others, sustaining **~3,230 deliveries/s** across **16 endpoints** — **2.7x** the ~1,202/s single-endpoint ceiling.
- Hardened against untrusted destinations with SSRF protection and AES-256-GCM-encrypted secrets at rest, verified by a `go test -race` suite covering crash recovery and idempotent duplicate-event handling.

link | **matscan** Sept 2026
Go

- Built a CLI-based matrix scanner for parsing and processing matrix input.

Achievements

- Topped CBSE Class XII 2020 Mathematics at Delhi Public School, Nacharam with **95%** marks

Education

Mahindra University Hyderabad
M.Tech in Artificial Intelligence & Data Science *2025 - Present*

- **Relevant Coursework:** Machine Learning, Data Science, Artificial Intelligence, Computer Vision

Mahindra University Hyderabad
B.Tech in Computer Science & Engineering *2021 - 2025*

- **Relevant Coursework:** Data Structures and Algorithms, Operating Systems, Database Management Systems, Computer Networks